BBHCSD Network Assessment and Troubleshooting Report

Remediation Recommendations Included

Provided by: TTG, LLC
Author: Michael Sokoloski
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Overview of BBHCSD network stated and observed issues.

- Devices kicked off network or losing connectivity on regular basis
- Time periods of unexplained latency
- Troubleshooting complexity unnecessarily high
- IP routing plan and vlans not ready for VoIP
- Overall network health and VoIP readiness is not in a good place

Testing, validation steps and discussions with network staff completed with observations.

- Ruckus network equipment validation completed, no equipment problems noted
    - Scale of devices
    - Sizing and ports
    - Interface sizing, error count checks, and utilization
    - Processor and utilization

- Layer 2 switching structures validated, implementation completed but overall network design problem
    - Spanning tree per vlan in good order
    - Leaf node path count high but not critical
    - Root bridge priority set and implemented correctly
    - Some vlans are spanned across all sites including WAN sites producing latency and collisions. Needs Corrected
    - Massive broadcast domains of end user devices. Needs Attention

- Layer 3 routing validated
    - Default routing noted through bridge devices then to Connect provider.
    - Once packet reaches Connect no further information is known
    - Only one building with layer 3 interfaces all buildings must span to reach for routing between vlans or outbound to Internet.
    - Routing design mixed with Connect provider

- Network security, web filtering and NAT
    - NAT (Network Address Translation) taking place at Connect for inbound and outbound Internet.
    - Web filter is single Barracuda device in bridge mode.
    - Firewall is single Linux iptables device in bridge mode.
    - No internal network security implemented
- Network Operations
    - Ad-hoc troubleshooting

- No monitoring of devices, interfaces or latency
- No alerting or event correlation
- No network operations run books

Remediation Recommendations with a focus on VoIP Readiness

Recommendation 1. (Must be completed before VoIP implementation)

- Breakup massive layer 2 (vlans) spanning all buildings including WAN sites.
  - Size of network has grown exponentially but network design approach has not kept pace
  - Use a campus design model
    - Each building needs a distribution layer with routing interfaces and trunked access layer switches.
    - Local IP traffic stays within the building
    - Vlans only span building or smaller logical unit if possible or warranted, greatly reducing broadcast domains
    - Intra-campus trunks become routing links for speed and isolation during troubleshooting.
    - Incoming services can be adopted quickly and put to best use.
    - Broadcast domains shrink producing significantly less noise for devices.
    - Security can be implemented in a controlled fashion
    - Operations and troubleshooting become significantly easier.
    - Wireless SSID can remain same but each building will have unique IP range.
    - WAN routing improvements and redundancy possible
    - Troubleshooting is improved by ability to isolate segments during troubleshooting process.
    - IP range schema developed for current and future expansion.
    - QoS settings become possible and valid

Recommendation 2. (Must be completed before VoIP implementation)

- Physical Cabling Plant
  - Current network has significant amount of workgroup style switches in offices, classes and labs.
  - This has a significant impact on the broadcast domain, and layer 2 structures in the network as each one of them is participating in the network. Typically, these are cheap throw away devices that can have sever negative impacts on the network as bugs in their code can introduce problems in the production network they are participating in.
  - Workgroups switches typically do not have PoE (power over ethernet) to power VoIP phones so power injectors would be needed.
  - Workgroup switches do not have the ability to hand off voice vlan access to VoIP devices which would impact QoS design and operation.
  - Troubleshooting is severely impacted by workgroup switch participation and there is the matter of power cords being unplugged.
  - A plan must be developed to find all workgroup switches in use and eliminate their usage by providing the correct amount of network drops direct from accessible IDFs.
  - This is being cared for in a separate SOW with NTE of $6,800.

Recommendation 3. (Critical to network operation and control but not strictly needed for VoIP)

- Network monitoring, alerting, and operations
    - Network monitoring, alerting and logging is needed for good network health and maintenance.
    - Network run books and other operational documents are needed for troubleshooting and improved performance.
    - Policy, procedures and responsibilities must be defined so that alerts and other critical data is not overlooked or not acted upon.

Recommendation 4. (Significant Risk but not strictly needed for VoIP)

- Design and implement increased network security in the following areas at a minimum.
    - Firewall and web filter redundancy and operation.
    - Network Security product selection, implementation and operation.
    - Wireless and guest access to the network.

Recommendation 5. (Longer term but carries risk not needed for VoIP)

- Specific IT needs for MDF and IDF rooms
    - HVAC
    - UPS and Generator
    - Physical security
    - Water penetrations and fire suppression

Assumption about VoIP server needs.

- Needs for virtual server capacity (memory, disk, processor) have taken place with district personnel if needed and any OS loads if not built from image or physical. If not, then this needs to be added to critical list.
- Rack space, power network port and other physical needs have been defined and signed off on with district personnel if needed. If not, then this needs to be added as well.

Implementation Options for your Consideration:

Recommendations 1 and 2 are critical prior to the roll out of voice on the network

Recommendation 3 should be implemented as well but not critical to VoIP success, however, definitely necessary for long term network success including VoIP

Optional SOW 1: TTG will provide the manpower and team necessary to complete the design elements of recommendation 1 and provide recommended subcontractors and bid costs for items to be implemented by a subcontractor (maybe Ttx) once the design phase is complete. The time frame from start to finish is estimated at 12 weeks assuming complete availability of district personnel, as needed, and the ability to implement changes by taking the networks down as needed. These outages will be announced, planned and communicated in advance and happen after 3:00 PM on weekdays or on weekends. TTG's fee for this scope of work would be $48,500.

Optional SOW 2: TTG to design and complete all work except for cable testing, which is already arranged and ready to roll. This will require TTG having complete access to the network firewalls and switches and assumes all the same as the above regarding availability of district personnel and ability to take the network down. The advantage of this SOW is that TTG will be contractor responsible for design and implementation. SOW 1 would involve an outside contractor (perhaps Ttx) for implementation. The potential downside here is it may take up to 20 weeks to complete this work. While we anticipate less 20 would be our commitment. This would put the phone system deployment off until Spring Break, most likely. TTG's fee for this scope of work would be $76,000.

The costs are based on either SOW 1 or 2 not both!!!

The real good news is there is no capital expense or no monies that have been misspent in the network upgrade process or VoIP implementation process. The redesign is simply necessary to upgrade the network routing and design like the district has already done with the network infrastructure hardware.

Optional SOW 3 - to be discussed as the project goes along is to rely on TTG for vCIO services to work with district staff on ongoing for implementation of recommendations 3, 4 and 5 to shore up the network in its entirety and keep it operating that way. Our vCIO packages are a small number of committed hours billed at a discounted rate depending on the hours. I believe we would see 20 to 30 hours per month for the first 12 months to handle the needs of the above outline of work and then probably 10 per month ongoing.

The below table represents our vCIO services and the associated rates. Rates are billed on a monthly ongoing basis based on your chosen package.

Block of Hours	Fees per Hour
0-10	$390
11-25	$350
26-50	$250
51-100	$190

This completes our recommendation for Brecksville school as it moves toward the implementation of its VoIP network infrastructure. It is very important to realize that BBHCSD already uses VoIP to transport calls between their building but with the new system will be using VoIP to carry calls within each building.

We are prepared to meet, review and discuss as you see necessary.